



Solar USB BICYCLE POWER BOX

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The circuit presented here can be used to drive USB-powered smart devices, including smartphones, while riding a bicycle. Smartphones suck battery power extremely fast, particularly when GPS and data connectivity are in constant use, and display and backlight are permanently powered up.

This circuit allows you to power up or recharge the battery of your USB gizmos with the help of solar power.

With 9V input from a small polycrystalline solar panel, this circuit produces a regulated output of 5V. The circuit can be used as a standalone 5V USB charger, or a 3.7V lithium-ion battery charger through the support of dedicated electronic components. Fig. 1 shows a typical solar USB bicycle power box.

Circuit and working

The circuit diagram of the solar USB bicycle power box is shown in Fig. 2. It is built around 9V, 450mA solar panel, two 1N5817 Schottky diodes (D1 and D2), AMS1117 5V fixed-voltage regulator (IC1), TP4056 battery charge controller (IC2) and a few other components.

The circuit is self-explanatory. DC voltage (9V@450mA) delivered by the solar panel connected to connector CON1 is routed to the low-drop fixed-voltage regulator AMS1117 (IC1). This input voltage is converted to USB-standard 5V DC supply, and is available through USB socket CON3.

Here, the two 1N5817 Schottky diodes (D1 and D2) are used in the input supply path as their forward voltage drop is only about half that of normal silicon diodes (approximately 0.3V rather than 0.75V per diode at 1A).

The 1000µF buffer capacitor (C1) is charged up to the maximum voltage delivered by the solar panel. For most 9V

solar panels, typical maximum output voltage is about 10V.

AMS1117 a 5V fixed-voltage regulator designed to provide up to 1A output current and to operate down to 1V input-to-output differential (dropout voltage of the device is guaranteed 1.3V maximum, decreasing at lower load currents). Note that, CON2 is multipurpose—it can take 9V DC supply catered by the solar panel, or it can feed external 9V DC supply to power the circuit when solar power is not available. Indicators LED1 and LED2 notify about the working status of the circuit.

Rest of the circuit is based on the dedicated lithium-ion TP4056 battery charge controller (IC2). TP4056 is a complete constant-current/constant-voltage linear charger for single-cell lithium-ion batteries. Other features include current monitor, under voltage lockout, automatic recharge, programmable charge current up to 1A, two status pins to indicate charge termination and presence of input voltage, etc.



Fig. 1: A typical solar USB bicycle power box

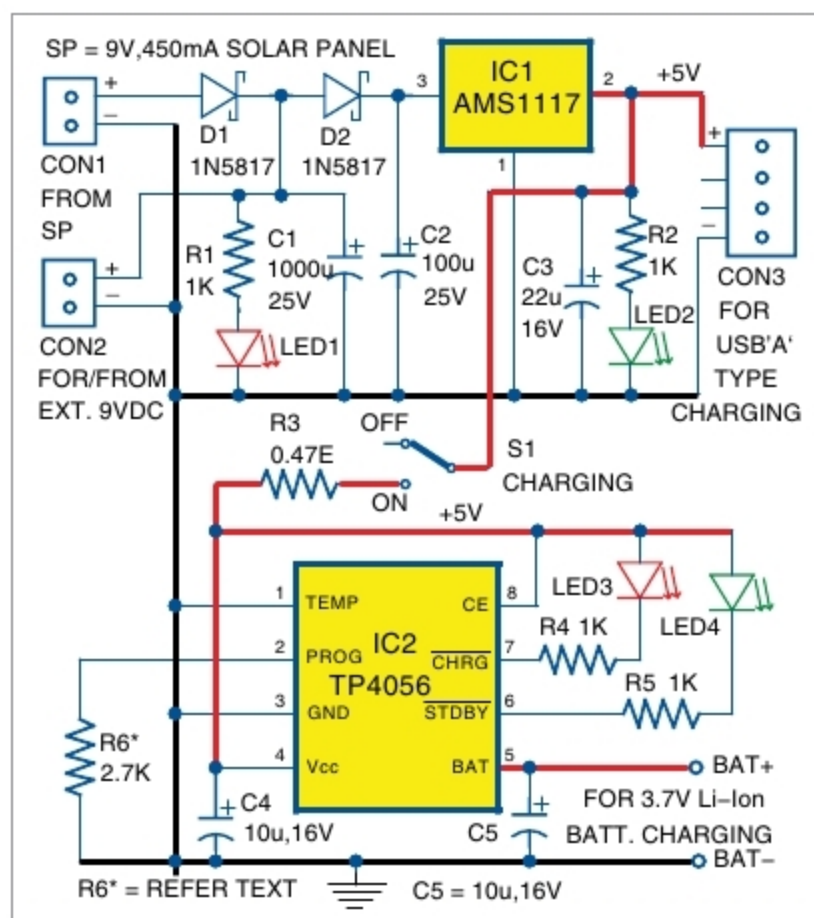


Fig. 2: Circuit diagram of solar USB bicycle power box

PARTS LIST

Semiconductors:

- IC1 - AMS1117, 5V fixed-voltage regulator
- IC2 - TP4056 battery charge controller
- D1, D2 - 1N5817 Schottky diode
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1, R2, R4, R5 - 1-kilo-ohm
- R3 - 0.47-ohm
- R6 - 2.7-kilo-ohm

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 100µF, 25V electrolytic
- C3 - 22µF, 16V electrolytic
- C4, C5 - 10µF, 16V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin connector
- CON3 - USB type-A connector
- S1 - SPDT switch
- BAT+, BAT- - Two connecting wires for the battery
- 9V, 450mA solar panel
- 3.7V lithium-ion battery for charging
- 9V DC (optional)